ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

GEOGRAPHY

PAPER 1: THEORY

MARKS: 100 **TIME: 2 hours**

Topics: the atmosphere, geomorphology, population, and water resources.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions. Answer ALL the questions.
2. Number your answers correctly. Write in full sentences where explanations are asked for.
3. Paragraph questions must be written in paragraph form, not in point form.

QUESTION 1: THE ATMOSPHERE

1.1 Choose the correct word(s) in brackets. Write only the word(s) next to the question number.

1.1.1 The layer of the atmosphere closest to the Earth's surface is the (troposphere / stratosphere). (1)

1.1.2 The ozone layer is found in the (stratosphere / mesosphere). (1)

1.1.3 The most abundant gas in the atmosphere is (oxygen / nitrogen). (1)

1.1.4 In the troposphere, temperature (increases / decreases) with height. (1)

1.1.5 The transfer of heat by rising and sinking air is called (conduction / convection). (1)

1.2 Dry air consists of about 78% nitrogen, 21% oxygen and 1% other gases.

1.2.1 Name the gas that makes up 21% of dry air. (1)

1.2.2 Name TWO greenhouse gases. (2)

1.2.3 Explain the greenhouse effect. (3)

1.2.4 Name TWO human activities that increase the amount of greenhouse gases. (2)

1.3 The ozone layer.

1.3.1 Why is the ozone layer important for life on Earth? (2)

1.3.2 Name the group of chemicals mainly responsible for damaging the ozone layer. (2)

1.3.3 Give TWO effects of ozone depletion on human health. (2)

1.4 In a paragraph of about SIX lines, explain how the sun heats the Earth's atmosphere. (6)

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QUESTION 2: GEOMORPHOLOGY

2.1 Give ONE geographical term for each description.

2.1.1 The thin, outermost solid layer of the Earth (1)

2.1.2 The thick layer of hot, partly molten rock below the crust (1)

2.1.3 The centre of the Earth, made mainly of iron and nickel (1)

2.1.4 The point on the Earth's surface directly above the focus of an earthquake (1)

2.1.5 Molten rock that reaches the Earth's surface (1)

2.2 Plate tectonics.

2.2.1 Name the THREE types of plate boundaries. (3)

2.2.2 At which type of plate boundary do fold mountains form? (1)

2.2.3 Name the theory that explains how the Earth's crust is divided into moving plates. (1)

2.2.4 Explain how convection currents in the mantle cause the plates to move. (2)

2.3 Earthquakes.

2.3.1 Define an earthquake. (2)

2.3.2 Name the scale used to measure the magnitude of an earthquake. (1)

2.3.3 Give TWO ways in which people can reduce the damage caused by earthquakes. (2)

2.4 Volcanoes.

2.4.1 Give TWO positive effects of volcanoes on people. (2)

2.4.2 Give TWO negative effects of volcanic eruptions on people. (2)

2.5 In a paragraph of about FOUR lines, explain why most earthquakes and volcanoes occur along plate boundaries, such as the Pacific "Ring of Fire". (4)

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QUESTION 3: POPULATION

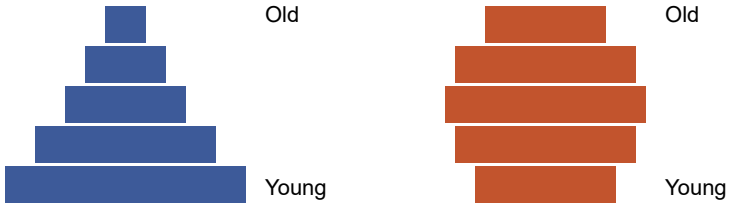
3.1 In a certain year, a country had a birth rate of 19 per 1 000 and a death rate of 9 per 1 000.

3.1.1 Define the term *population growth rate*. (2)

3.1.2 Calculate the natural increase of the population as a percentage. Show your working. (3)

3.1.3 What term describes the number of people living per square kilometre? (1)

3.2 Study the two simplified population pyramids below.



Country A **Country B**

3.2.1 Which country (A or B) is most likely a developing country? (1)

3.2.2 Give a reason for your answer in QUESTION 3.2.1. (2)

3.2.3 Give TWO problems faced by a country with a very young population. (2)

3.2.4 Give TWO problems faced by a country with an ageing population. (2)

3.3 Migration.

3.3.1 Define *migration*. (2)

3.3.2 Give TWO push factors that make people leave rural areas. (2)

3.3.3 Give TWO pull factors that attract people to cities. (2)

3.3.4 Explain what is meant by a *brain drain*. (2)

3.4 In a paragraph of about FOUR lines, explain how rapid population growth puts pressure on resources such as water, schools and housing. (4)

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QUESTION 4: WATER RESOURCES

4.1 Ocean currents around South Africa.

4.1.1 Name the warm ocean current along the east coast of South Africa. (1)

4.1.2 Name the cold ocean current along the west coast of South Africa. (1)

4.1.3 Explain how the cold current influences the climate of the west coast. (2)

4.2 Floods and droughts.

4.2.1 Define *drought*. (2)

4.2.2 Give TWO causes of floods. (2)

4.2.3 Give TWO ways in which flood damage can be reduced. (2)

4.2.4 Explain ONE way in which a drought affects farmers. (2)

4.3 Water management in South Africa.

4.3.1 Explain why South Africa is described as a water-scarce country. (2)

4.3.2 Name ONE large project that transfers water to Gauteng from another country. (1)

4.3.3 Give TWO ways in which households can save water. (2)

4.3.4 Explain why water pollution is a serious problem in South Africa. (2)

4.4 In a paragraph of about SIX lines, explain how climate change could affect the supply of water in South Africa. (6)

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TOTAL: 100

END OF QUESTION PAPER